

Types of Data Structures: Linear & Non-Linear

Q. Classify Data Structures into Linear and Non-Linear types. Differentiate between them and provide examples of each.

[DEF] Definition to Remember

A Data Structure is a specialized format for organizing, processing, retrieving, and storing data. Based on how elements are arranged, data structures are classified into Linear (elements in sequential order) and Non-Linear (elements arranged hierarchically or in a network).

1. Linear Data Structures

- * Elements are arranged in sequential order -- each element has a unique predecessor and successor.

Examples:

DS	Description
Array	Contiguous memory block, same data typ
Linked List	Chain of nodes; each node points to th
Stack	Follows LIFO (Last In, First Out)
Queue	Follows FIFO (First In, First Out)

2. Non-Linear Data Structures

- * Elements are arranged in a hierarchical or network structure.

Examples:

DS	Description
Tree	Hierarchical structure with a root nod
Graph	Network of vertices (nodes) and

3. Linear vs Non-Linear Comparison

Feature	Linear	Non-Linear
Arrangement	Sequential order	Hierarchical / network
Traversal	Single run	Multiple paths / recursion
Memory	Often less efficient	More efficient (dynamic allocation)
Implementation	Simpler	More complex
Levels	Single level	Multi-level
Examples	Array, Stack, Queue, Linked List	Tree, Graph

LINEAR: [A] -> [B] -> [C] -> [D]

NON-LINEAR (Tree):

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  [A]
 /  \
[B]  [C]
 /  \
[D]  [E]
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[MUST-WRITE] Must-Write Points (for 10 marks)

1. Data structures are classified into Linear (sequential) and Non-Linear (hierarchical/network).
2. Linear DS: elements arranged sequentially -- each has one predecessor and one successor.
3. Linear examples: Array, Linked List, Stack (LIFO), Queue (FIFO).
4. Non-Linear DS: elements connected to multiple others -- Tree and Graph.
5. Linear DS can be traversed in a single run; Non-Linear requires multiple paths.
6. Non-Linear DS uses memory more efficiently via dynamic allocation.
7. Trees are used in database indexing; Graphs are used in route planning and AI.

[RECALL] Quick Recall

Linear (sequential, single run) -> Array, Linked List, Stack (LIFO), Queue (FIFO) -> Non-Linear (hierarchical/network, multi-path) -> Tree, Graph -> Non-Linear: efficient memory, complex traversal